香 港 大 學

THE UNIVERSITY OF HONG KONG

COMP4801 Final Year Project
Detailed Project Plan

Aggregating Epigenetic Methylation Signals for Improved Long-Read Genetic Variant Calling

Name: LI Junzhe

UID: 3035973854

Supervisor: Prof. Luo Ruibang

Date: October 1, 2025

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1. Project Background

1.1 The Critical Role and Challenges of Genomic Variant Calling

The accurate identification of genomic variants, which are differences in DNA sequence relative to a reference genome, represents a foundational pillar of modern genomics, clinical diagnostics, and biomedical research. These variants encompass a wide spectrum, from single nucleotide polymorphisms (SNPs) and small insertions/deletions (InDels) to large-scale structural variations (SVs) . Their analysis is critical for understanding human genetic diversity, diagnosing Mendelian disorders, developing targeted cancer therapies, and assessing disease risk [1]. The proliferation of Next-Generation Sequencing (NGS) technologies has profoundly enhanced our capacity to detect these variants at a genome-wide scale [2].

Despite these technological advances, the process of variant calling remains a significant bioinformatic challenge. Even state-of-the-art computational pipelines frequently yield discordant results, suffering from non-negligible rates of both false positive and false negative calls [1]. These inaccuracies are not uniformly distributed across the genome; they are systematically enriched in regions characterized by low sequence complexity, high repetitiveness (such as tandem repeats or segmental duplications), or extensive homozygosity [3]. In these "difficult-to-map" regions, the alignment of sequencing reads becomes ambiguous. A read originating from a repetitive element, for instance, may map with equal confidence to multiple locations in the reference genome. Consequently, current variant calling methods are forced into a compromise: they either produce erroneous calls due to mapping artifacts or, more commonly, they sacrifice detection sensitivity in these regions to maintain an acceptable false discovery rate, effectively leaving significant portions of the genome unanalyzed [1].

1.2 The Rise of Long-Read Sequencing and Haplotype-Aware Methods

Long-read sequencing platforms, led by Pacific Biosciences (PacBio) and Oxford Nanopore Technologies (ONT), offer a powerful solution to many of the limitations

imposed by short-read technologies [4]. By generating reads that can be tens to hundreds of thousands of base pairs in length, these technologies can span entire repetitive elements and provide the long-range contextual information required to resolve complex genomic architectures and unambiguously map reads [4].

This capability has spurred the development of a new class of sophisticated "haplotype-aware" variant callers, including the GATK HaplotypeCaller and, more recently, pipelines like Clair3 or PEPPER-Margin-DeepVariant designed specifically for long reads [5]. Rather than assessing each genomic position independently, these tools operate on a more biologically faithful principle. In a given region of interest, they perform a local *de novo* assembly of the sequencing reads to reconstruct the two parental chromosome copies, or haplotypes, present in a diploid organism. By evaluating how well each read supports the reconstructed haplotypes, these methods can more effectively distinguish true genetic variants from random sequencing errors and systematically call SNPs and indels simultaneously [5].

1.3 The Core Limitation: Dependency on SNP-Based Phasing

While powerful, the efficacy of virtually all contemporary haplotype-aware callers is fundamentally predicated on a single type of genetic marker: the heterozygous SNP. The process of "phasing"—the critical first step of partitioning sequencing reads into two parent-of-origin bins—relies on identifying reads that cover these known heterozygous sites.

This dependency creates a critical algorithmic vulnerability. In genomic regions where heterozygous SNPs are sparse or entirely absent—a defining characteristic of homozygous regions, regions of low mappability, or certain repetitive elements—the phasing algorithm is deprived of its necessary input [6]. This leads to a cascade of negative consequences: phase blocks become fragmented and short, reads that do not span a heterozygous site cannot be assigned a haplotype, and the overall power of the haplotype-aware model is severely diminished. Consequently, variant calling accuracy and completeness plummet precisely in the genomic contexts that are already the most

challenging to analyze. The central problem is a causal link between local SNP density and the performance of the phasing algorithm. This project introduces an orthogonal data type—DNA methylation—that is not subject to the same constraint. Allele-Specific Methylation can serve as a robust haplotype marker even in the absence of nearby heterozygous SNPs [7], thereby decoupling the phasing process from the underlying genetic landscape and offering a new paradigm for analysis, similar in concept to recent methods that also leverage methylation for phasing [6].

2. Project Objective

The primary objective of this project is to design, implement, and rigorously validate a novel computational method that integrates Allele-Specific Methylation (ASM) signals, natively detected in modern long-read sequencing [8], to enhance the accuracy, completeness, and contiguity of haplotype-aware variant calling.

The specific, measurable objectives are as follows:

2.1 Identify ASM Sites from ONT Long-read Sequencing Data

This project aims to develop a robust algorithm to statistically identify ASM sites within existing phase sets from ONT long-read sequencing data that contains simultaneous sequence and methylation information, which is stored in BAM files as MM and ML tags [9]. 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC) are taken into account [10][11].

2.2 Increased Phasing Performance

This project aims to implement a method to leverage these identified ASM "epigenetic markers" to assign definitive haplotype identity to sequencing reads that were previously unphased due to a lack of spanning SNP information. This will increase the percentage of haplotagged reads. We will also create a module that concatenates adjacent, but previously disconnected, phase sets based on consistent ASM patterns observed in overlapping reads, thereby increasing the length and contiguity of

reconstructed haplotypes. Better phasing performance can be observed in Binary Alignment Map (BAM) files.

2.3 Better Variant Calling Performance

This project will focus on the improvement over a baseline SNP-only phasing method across a suite of key metrics, including VCF Quality Score (QUAL), precision, recall, and F1-score for both SNPs and overall variants.

3. Project Methodology

This project will be implemented as a bioinformatics pipeline primarily written in Python, leveraging established scientific libraries such as pysam for efficient manipulation of sequencing alignment files, scipy for statistical computations, and pandas for data management. It will be integrated into the advanced Clair3 variant calling workflow [12]. The methodology section is structured into three principal stages: Data Preparation, Algorithm Design and Evaluation.

3.1 Data Acquisition and Preparation

The primary dataset for the development, testing, and evaluation will be the publicly available Genome in a Bottle (GIAB) whole-genome sequencing dataset [13]. This dataset is the industry standard for benchmarking variant calling methods, as it is accompanied by a high-confidence "truth" variant call set against which performance can be accurately measured. Crucially, we will utilize data generated with Oxford Nanopore Technology, which provides both long, highly accurate reads and simultaneous, base-resolution detection of 5mC and 5hmc methylation in its native state, without the need for destructive chemical treatments like bisulfite conversion, To assess the robustness of the method, datasets at various sequencing coverages (e.g., 10x, 20x, 30x, 40x, 50x) will be analyzed. The initial data preparation step will involve aligning the raw reads to the GRCh38 human reference genome using a long-read aligner to produce the primary input for the pipeline: a coordinate-sorted BAM file containing

methylation tags.

3.2 Algorithm Implementation

The algorithm recursively identifies ASM sites within phase sets using Fisher's Exact Test and uses these epigenetic markers to assign haplotypes to previously unphased reads [11]. The method also concatenates adjacent phase sets, enhancing ASM phasing reliability and extending haplotype length.

3.2.1 Interpreting the Input

The algorithm's input is a BAM file generated by a long-read sequencer, where the reads have already been haplotagged by a standard tool (e.g., WhatsHap). This file will contain reads assigned to Haplotype 1 (HP:i:1), Haplotype 2 (HP:i:2), and a subset of unphased reads (lacking an HP tag). Methylation information are stored in MM and ML tags in the BAM file. MM shows the location of the 5mc and 5hmc, and ML is a 0-255 number showing the confidence that a 5mc or 5hmc exists. ML is then translated into the 0 or 1 signals by using Gini Impurity to find out the best threshold [11].

3.2.2 ASM Site Identification

The algorithm identifies ASM Site through each existing phase set (a contiguous genomic region with a group of phased reads). Within each set, it systematically scans for CpG dinucleotides that exhibit a bimodal methylation pattern—that is, sites where a substantial fraction of covering reads are methylated and a substantial fraction are unmethylated [14]. For each such candidate site, a 2x2 contingency table is constructed:

	Methylated Reads	Unmethylated Reads
Haplotype 1	Count A	Count B
Haplotype 2	Count C	Count D

Fisher's Exact Test is then applied to this table. This statistical test is chosen for its

accuracy in analyzing categorical data, especially when cell counts may be small, as is common in sequencing data. The test determines if there is a statistically significant non-random association between haplotype assignment and methylation status (e.g., p-value < 0.05). A significant result validates the site as a high-confidence ASM locus, where methylation status acts as a reliable proxy for haplotype identity.

3.2.3 Epigenetic Haplotype Assignment

After identifying a catalog of high-confidence ASM sites, the algorithm performs a second pass over all previously unphased reads that cover these loci. If an unphased read displays a methylation pattern that is strongly characteristic of one of the defined haplotypes (e.g., the read is methylated at a site where Haplotype 1 is predominantly methylated and Haplotype 2 is not), it is confidently assigned to that haplotype by adding the appropriate HP tag. This step directly recovers phasing information that was lost by SNP-only methods.

3.2.4 Phase Set Concatenation

Finally, the algorithm inspects the boundaries of adjacent but disconnected phase sets. If two neighboring sets are found to share a consistent ASM signature on reads that span the gap between them (e.g., reads assigned to Haplotype 1 are methylated in both sets at a linking ASM site), the algorithm merges them into a single, longer, and more contiguous phase block. This directly addresses the critical goal of increasing overall haplotype length and contiguity.

3.3 Evaluation and Benchmarking

The output of our algorithm will be a phased BAM files. The final output of the integrated pipeline will be a variant call file (VCF). The accuracy of this VCF will be rigorously benchmarked against the GIAB high-confidence truth set using standard evaluation tools like hap.py. The evaluation will be multifaceted, focusing on two distinct but related areas of improvement:

1. **Phasing Improvement:** The primary metrics will be the absolute and percentage increase in the number of reads assigned an HP tag in the BAM file, demonstrating the direct recovery of phasing information. Furthermore, we will measure the increase in haplotype contiguity using the phase block N50 statistic, a standard measure of assembly continuity.
2. **Variant Calling Accuracy:** Standard performance metrics—precision, recall, and F1-score—will be calculated for both SNPs and indels. The VCF QUAL score, a logarithmic measure of call confidence, will also be compared. A key component of the analysis will be to stratify these results by genomic context to demonstrate that the most significant improvements occur in regions known to be challenging for standard methods.

This evaluation strategy is designed not only to measure improvements in point accuracy (i.e., correctly identifying individual variants) but also to capture improvements in the structural fidelity of the reconstructed haplotypes. Increasing phase block length provides more definitive information about whether two variants are on the same parental chromosome (*cis*) or on different ones (*trans*). This long-range phasing information is often more biologically and clinically significant than a minor increase in the F1-score for individual variant calls, as it is essential for understanding compound heterozygosity and complex gene regulation.

4. Project Schedule and Milestones

The project is organized into three sequential phases, which are aligned with the specified deliverable deadlines. The detailed schedule, tasks, and milestones are outlined in the chart below. Deliverables are highlighted in boldface.

Time	Milestones
Phase 1	

Time	Milestones
Sep 15, 2025	Research on Background Information
Sep 20, 2025	Finalize Project Scope & Methodology Prepare Development Environment & Data
Oct 1, 2025	**Detailed Project Plan**
Oct 1, 2025	**Project Web Page Setup**
Phase 2	
Nov 30, 2025	Carefully Examine the Data Attributes and Design Algorithm to Process the Methylation Data
Dec 20, 2025	Implement Core ASM Detection Module, Haplotype Assignment Logic and Phase Set Concatenation Logic
Jan 10, 2026	Preliminary results, Initial Testing on GIAB Subset
Jan 12, 2026	**First Presentation**
Jan 25, 2026	Preliminary Implementation **Detailed Interim Report**
Phase 3	
Feb 15, 2026	Finalized algorithm Perform ablation study
Mar 15, 2026	Perform Full-Scale Benchmarking Prepare Raw benchmark data, plots

Time	Milestones
Mar 31, 2026	Results Analysis & Interpretation Prepare Tables and Figures for Report
Apr 19, 2026	**Finalized Tested Implementation & Final Report**
Apr 19, 2026	**Project Web Page (Final Version)**
Apr 22, 2026	**Final Presentation**
Apr 25, 2026	**Poster, 1-min Video**

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